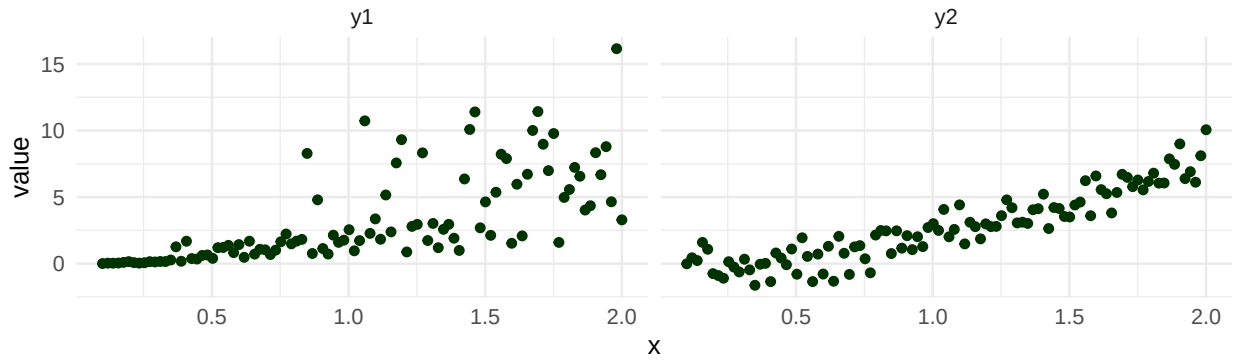
Day 08: Transformations and polynomial regression

Warm up: Consider the following two models:

- $\log_2(y) = 1 + 2\log_2(x) + \varepsilon,\ \varepsilon \sim N(0, 1)$
- $y = 2x^2 + \varepsilon,\ \varepsilon \sim N(0, 1)$

Show that the two models have the same $\hat{y}$'s.

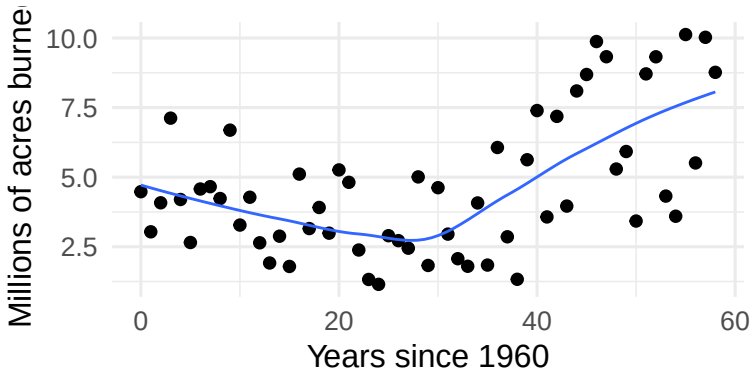
> Hint: log rules might be useful
>
> - $\log(ab) = \log(a) + \log(b)$
> - $\log(a^b) = b\log(a)$
> - $\log(a/b) = \log(a) - \log(b)$
> - $\log_b b = 1$



Even though they give the same **predictions**, Model (2) has constant variance but Model (1) does not. If we use Model (2) to make prediction intervals for data that actually look like Model (1), our interval will be too wide for small x and too narrow for large x.

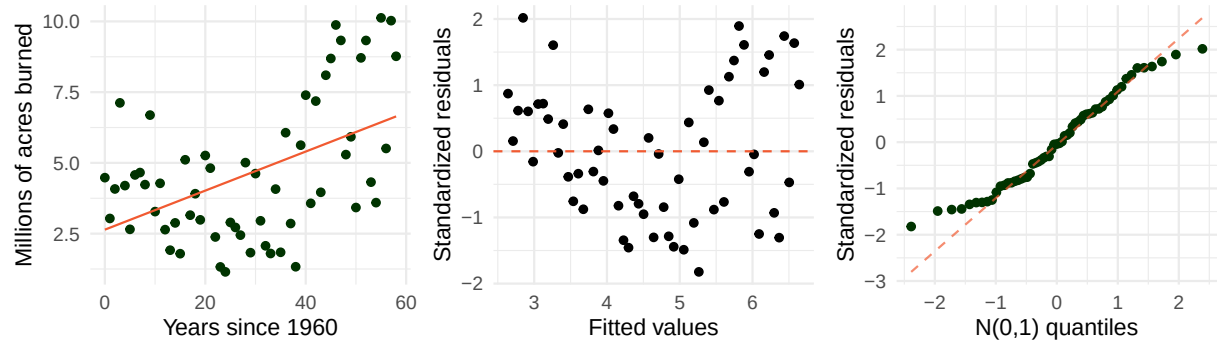
Data for today: wildfires

The National Interagency Coordination Center compiles annual wildland fire statistics for federal and state agencies (Situation Reports, used for several decades). Our goal is to model the number of acres burned over the years.

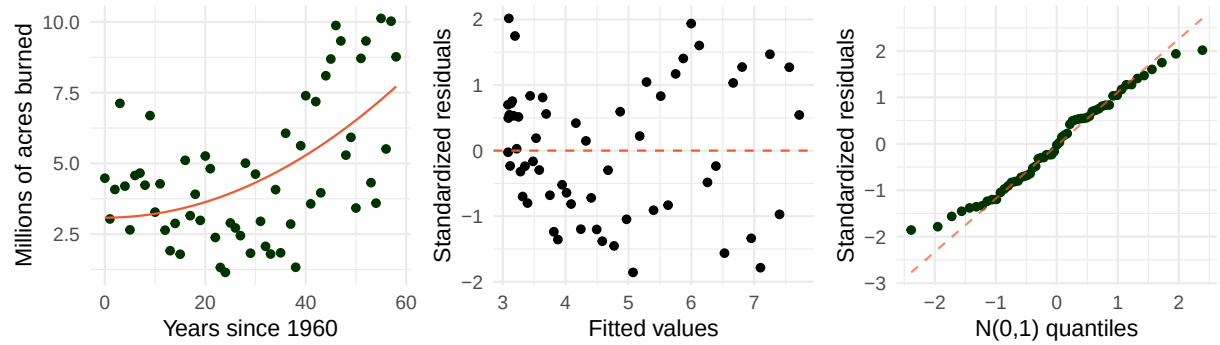


Three candidate mean functions

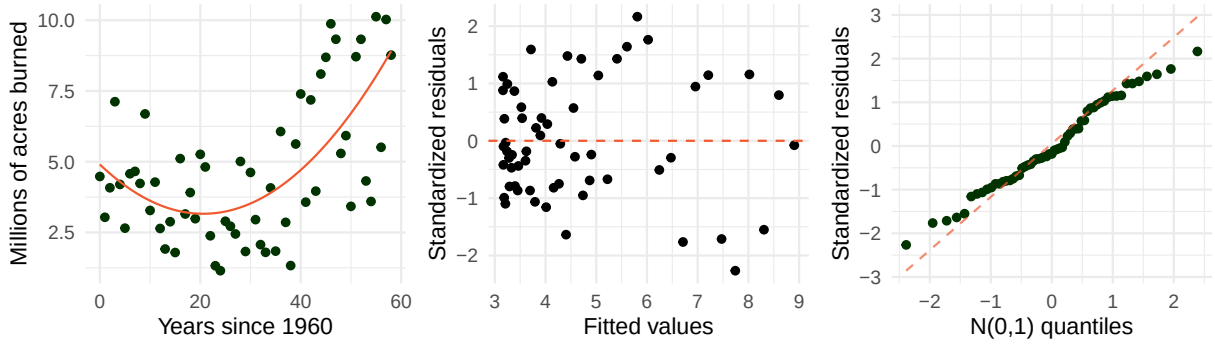
Option 1: SLR $\mu\{y \mid x\} = \beta_0 + \beta_1 x$



Option 2: transform X $\mu\{y \mid x\} = \beta_0 + \beta_1 x^2$



Option 3: polynomial $\mu\{y \mid x\} = \beta_0 + \beta_1 x + \beta_2 x^2$



The polynomial regression model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_k x_i^k + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma)$$

Assumptions (same as in SLR):

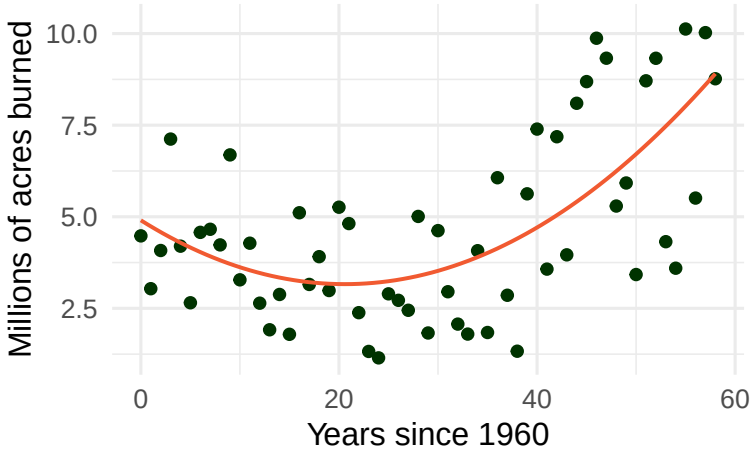
1. $\mu\{Y \mid x_i\}$ is a linear function of the _____
2. I:
3. For each x_i, the sub-population of responses is normally distributed
4. The standard deviation for each sub-population is _____

Interpreting the model coefficients

With a quadratic term, we can no longer talk about "the" slope. Instead we focus on the expected change in y for a _____ increase in x.

$\mu\{y \mid x+1\} - \mu\{y \mid x\} =$

Notice this expected change **depends on** _____



Inference about coefficients

Inference uses the same t-based tools as SLR, but with

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{n} e_i^2}{n-(k+1)}}$$

and the degrees of freedom of the reference t-distribution change to $n-(k+1)$, where k is:

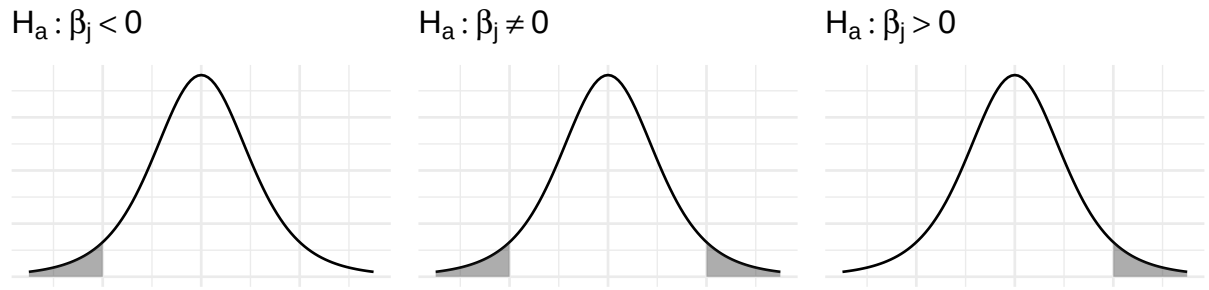
Testing a single coefficient

Hypotheses: $H_0 : \beta_j = \#$ vs. $H_a : \beta_j \begin{matrix} < \\ \neq \\ > \end{matrix}$ null value

Test statistic: $t = \dfrac{\hat{\beta}_j - \text{null value}}{SE(\hat{\beta}_j)}$

Reference distribution: t distribution with $df = n-(k+1)$

p-value: area in the tail(s) specified by H_a



Example: Wildfires

term	estimate	std. error	t	p-value
Intercept	4.903	0.721	6.80	<0.001
Year	-0.169	0.058	-2.94	0.005
Year^2	0.004	0.001	4.28	<0.001

Try it

Would a higher-order polynomial (e.g. cubic, quartic, quintic) provide a *better* fit to the wildfire data? Work through the R handout with your group.

1. Does the cubic model appear to be necessary? Use the `summary()` function to explore the fitted model and run a hypothesis test for the cubic term. What do you conclude?

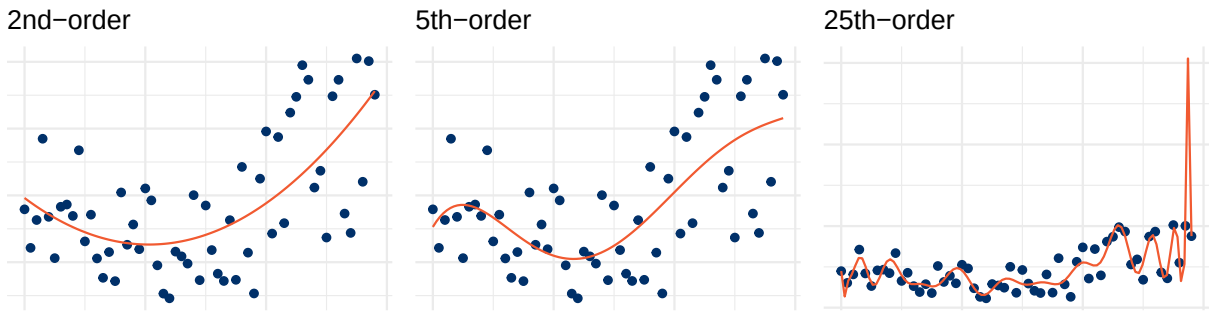
2. What degrees of freedom did R use for the t-distribution used to calculate the p-value for the test of the cubic term?

3. Do you notice anything curious about the inferential results for the linear and quadratic terms?

> **!** Stop here
>
> Let Amanda know if/when your group finishes the R activity

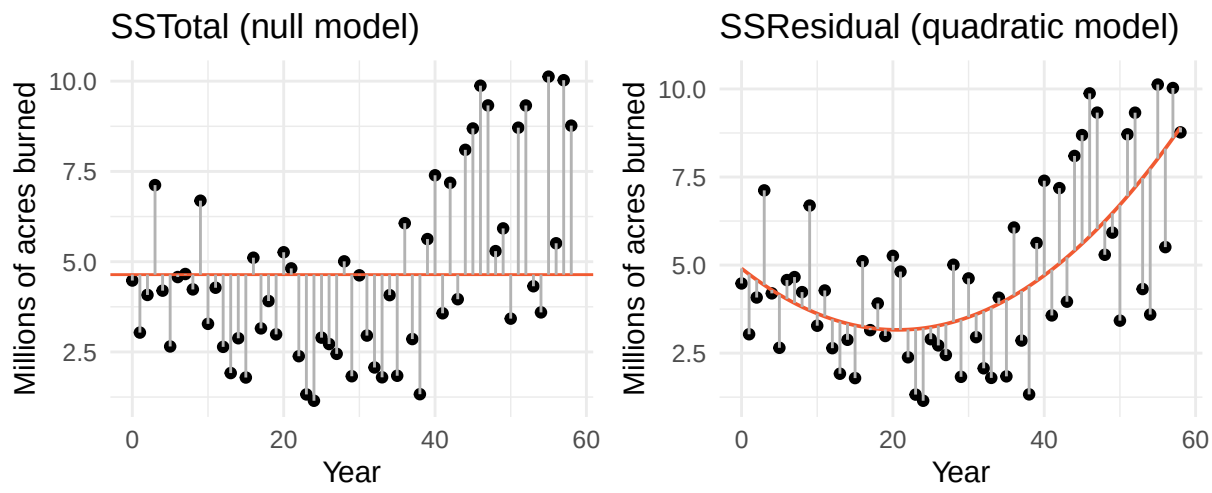
A warning about polynomials

High-order polynomial regression models will over-fit your data rather than capturing the true, generalizable trend.



Decomposing the variability

- **Null model** ($\mu\{Y \mid X\} = \beta_0$): uses $\overline{y}$ as the prediction for every point
- SSTotal $= \sum(Y_i - \overline{Y})^2$ – overall variability in Y.
- SSResidual $= \sum(Y_i - \hat{Y}_i)^2$ – variability left *unexplained* by the model - *Note:* $\hat{\sigma} = \sqrt{\text{SSResidual}/df}$.
- SSRegression $=$ SSTotal $-$ SSResidual is the variability *explained* by the model.



Coefficient of determination (R^2)

Proportion of the total variation in y explained by the model:

$$R^2 = \frac{\text{SSRegression}}{\text{SSTotal}} = 1 - \frac{\text{SSResidual}}{\text{SSTotal}}$$